Sectorized Observable Framework

A Typed Static Object Language with Exact Marked Finite Realizations

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This paper is Paper VIII of the RIME program. It begins the static typed object layer by identifying the data on which sectorized observable constructions are defined. Papers IV–VII provide compatible earlier interfaces but remain logically independent.

Abstract

Problem. Sectorized analyses repeatedly use projectors, observable families, projected blocks, routed products, words, commutators, depth filtrations, and several distinct operator-algebra closures. Treating these objects as a single accessibility ladder obscures the carrier, labels, word length, and closure conventions that determine their mathematical meaning.
Approach. We define a finite Sectorized Observable Framework (SOF) as a marked sector realization together with a labelled observable map $Y : A \to \mathcal{B}(V)$, which need not be injective. The static data include the marked sector algebra D_Q, the labelled family Y, the observable operator system E_Y, and the sector-enriched system $S_{Q,Y}$. We separate the positive word algebra A_Y^+, the observable star-closure A_Y^*, and the sector-enriched star-closure $A_{Q,Y}^*$. We then define separate operator/word and Lie/Hall carriers, their strict morphisms, and their functorial constructions.
Results. A declared realization supplies a well-defined operator SOF core. Strict operator morphisms form a category, as do independently enriched Lie/Hall morphisms. Exact algebra or $*$-isomorphisms identify the matched multiplication closures. Two functoriality theorems map labelled operator blocks, routed products, full words, and registered Lie/Hall witnesses forward, yielding non-increasing target depths. Equality is asserted under strict equivalence, not under an arbitrary strict embedding. The two branches are not identified without an explicit bridge theorem. Positive-word, star-word, and sector-enriched saturated corners are recorded separately from finite first-hit filtrations. As an exact realization family, a finite permutation action together with a declared label map and marked state partition produces a coordinate-sector SOF. In this family routed and full-word support coincide, while aggregate Boolean paths can still strictly overestimate them. Finite represented-image saturation gives exact first-hit word depths, including certified nonreachability when present. A paper-owned conformance certificate replays six modular actions, seventeen bounded triangle-group actions, fifty-one marked sectorizations, and a four-state strict path/word witness.
Boundary. The sectorization may be representation-derived, geometry-derived, filtration-derived, activation-derived, or externally chosen. The realization construction is formal relative to the declared choices, not a classification theorem and not a proof that every source has a unique or canonical SOF realization. A finite representation alone is not sufficient data for the labelled operative map or marked partition. The promoted finite family supplies no Lie/Hall carrier and does not establish generic strict separation of positive and star closures. This paper does not claim a complete weak deformation category, moving accessibility fields, universal completion, or an unconditional relation between word and Lie depth.

Notation and Claim Layers

Symbol	Meaning
V	finite-dimensional complex Hilbert space
Q_i	orthogonal sector projector
$D_Q = \mathrm{span}_{\mathbb{C}}\{Q_i\}$	marked sector algebra
$Y : A \to \mathcal{B}(V)$	declared labelled operator and word alphabet; the map need not be injective
E_Y	$\mathrm{span}_{\mathbb{C}}\{I, Y_a, Y_a^* : a \in A\}$
$S_{Q,Y}$	$D_Q + E_Y$, the sector-enriched operator system
A_Y^+	$\mathrm{alg}_{\mathbb{C}}(I, Y)$, the positive associative word algebra
A_Y^*	$C^*(Y)$, the observable star-closure
$A_{Q,Y}^*$	$C^*(D_Q \cup Y)$, the sector-enriched star-closure
B_{ij}^a	$Q_i Y_a Q_j$, a labelled operator block
$R_1[Y]$	aggregate direct support of the labelled operator family
$\mathcal{R}_{d,ij}[Y]$	linear span of length-d routed products from j to i
$\mathrm{Route}_d[Y]$	support of nonzero routed projected products
$\mathcal{W}_d(Y)$	linear span of full ordered words of length d
$W_d[Y]$	support of nonzero full ordered words of length d
$D_{\mathrm{route}}[Y]$	first routed-product depth
$D_{\mathrm{word}}[Y]$	first full-word depth
X	independently registered skew-adjoint Lie family
$R_1^{\mathrm{Lie}}, R_2^{\mathrm{Lie}}$	direct and simple-commutator Lie shadows
D_{Lie}	depth relative to a declared Hall/Lie filtration
$G \curvearrowright \Omega$	finite action used by the exact marked-permutation realization family
$\Pi = \{\Omega_i\}_{i \in I}$	declared marked partition of the finite state set
$C_{\leq d}[Y]$	represented operators reached by positive words of lengths 1 through d

The four reader-facing evidence levels are:

1. **Theorem:** an exact result proved from declared hypotheses;
2. **Computational Certificate:** a reproducible finite computation tied to declared inputs;
3. **Computational Observation:** a bounded numerical pattern without promotion;
4. **Research Program:** an open problem, conjectural bridge, or proposed extension.

Definitions are not evidence claims. Arrows in object diagrams denote construction or audit order unless a theorem explicitly states an implication.

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1 Introduction

1.1 Problem and Scope

The object studied here is not a particular representation, group, or physical model. Instead, it is a finite sectorized observable realization: a space, a compatible sectorization, and a declared observable alphabet. The sectorization is source-dependent. It may arise from representation theory, a Dirac operator, a mesh or interface geometry, a state filtration, a graph coloring, an activation rule, a control decomposition, or an external coarse coordinate choice. The observable language begins after that choice has been made.

The static object language must preserve distinctions that are frequently obscured by the shorthand

$$R_1 \longrightarrow R_2 \longrightarrow D.$$

In particular, a labelled operator family is not the same as its linear span, an operator system is not a word filtration, a routed product is not a full word, and a simple commutator is not a second associative word layer. Positive multiplication closure, adjoint closure, and closure under internal sector markers are also distinct operations. A saturated algebra records closure, not the first length at which a sector pair is reached.

This paper therefore has five aims:

1. define the marked static SOF data;
2. define typed operator/word and Lie/Hall carriers;
3. define strict static morphisms and their categories;
4. prove carrier-qualified support preservation and depth monotonicity;
5. provide an exact marked finite-permutation realization family with a source-addressed conformance certificate and strict Boolean-path control.

This is a realization construction, not a classification theorem. It says what data are sufficient to enter the SOF object language; it does not say that all source systems admit the same canonical sectorization.

1.2 Relation to Earlier Work

The fixed spectral arrangements of Paper IV, the typed fixed-family objects of Paper V, the normality-gated point registrations of Paper VI, and the projected-composition incidence geometry of Paper VII supply compatible interfaces for the present static language [2, 1, 4, 3]. They remain independent papers. Paper VI is not used here as a positive moving SOF instance: the cited evidence consists of linearized commutativity/normality certificates and pointwise registrations, while coherent moving projector fields remain open.

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2 Static SOF Objects

2.1 Definition (Operator SOF)

Let V be a finite-dimensional complex Hilbert space and let $\{Q_i\}_{i\in I}$ be a finite complete orthogonal sectorization:

$$Q_i^* = Q_i, \qquad Q_i \neq 0, \qquad Q_iQ_j = \delta_{ij}Q_i, \qquad \sum_{i\in I} Q_i = I.$$

Here compatibility means that the projectors and declared observables act on the same space and that any source-specific admissibility conditions have been recorded. It does not mean that the Q_i commute with, or reduce, the declared observables.

Let A_0 be a finite nonempty label set and let

$$Y_0 : A_0 \longrightarrow \mathcal{B}(V), \qquad a \longmapsto Y_a^{(0)},$$

be an extracted labelled operator map. It is not required to be injective: distinct source labels may have the same represented operator. Its operative word convention is fixed before the SOF core is declared. A positive-word convention sets $Y = Y_0$. A star-word convention sets $A = A_0 \times \{+, -\}$ and declares the labelled completion

$$Y^{\mathrm{adj}} = \{Y_{(a,+)} = Y_a^{(0)},\ Y_{(a,-)} = (Y_a^{(0)})^* : a \in A_0\},$$

with $Y = Y^{\mathrm{adj}}$. Distinct completion labels remain distinct even when their represented operators agree. After this registration, $Y : A \to \mathcal{B}(V)$ denotes the selected operative alphabet in all direct-support, routed-product, path, word, and depth constructions below. Sector projectors are not letters of Y unless a realization explicitly declares them as observables. An **operator SOF core** is the tuple

$$\mathcal{F}_{\mathrm{op}} = (V, \{Q_i\}_{i \in I}, Y : A \to \mathcal{B}(V)).$$

The possibly noninjective label map $a \mapsto Y_a$, including any adjoint-completion labels, is part of the data. Passing to its image set, a span, or a generated algebra may forget labels, multiplicities, order, and word length; therefore those constructions are recorded separately.

The marked sector algebra, observable operator system, and sector-enriched operator system are

$$D_Q = \mathrm{span}_{\mathbb{C}}\{Q_i : i \in I\},$$

$$E_Y = \mathrm{span}_{\mathbb{C}}\{I, Y_a, Y_a^* : a \in A\},$$

$$S_{Q,Y} = D_Q + E_Y.$$

E_Y and $S_{Q,Y}$ carry their standard operator-system meanings. Here SOF abbreviates Sectorized Observable Framework, not operator system.

Three multiplication closures are recorded separately. The **positive associative word algebra** is

$$A_Y^+ := \mathrm{alg}_{\mathbb{C}}(I, Y).$$

It uses finite products of the declared operative letters and does not add undeclared adjoints. It need not be adjoint-closed or semisimple. The **observable star-closure** is

$$A_Y^* := C^*(Y),$$

the smallest unital $*$-subalgebra of $\mathcal{B}(V)$ containing the declared observables. The **sector-enriched star-closure** is

$$A_{Q,Y}^* := C^*(D_Q \cup Y) = C^*(S_{Q,Y}).$$

The last closure permits sector projectors to occur as internal multiplicative separators. It therefore contains routed star-words, not only full words in the operative alphabet. If the operative alphabet is already adjoint-completed, then $A_Y^+ = A_Y^*$; this equality is a consequence of that declared convention, not a default identification.

Finite-dimensional C^*-algebras are semisimple by standard theory, so the two star-closures have Wedderburn decompositions. That background fact does not automatically apply to A_Y^+, nor does it constitute an SOF theorem. The marked tuple

$$(D_Q, Y, E_Y, S_{Q,Y}, A_Y^+, A_Y^*, A_{Q,Y}^*)$$

retains information that the abstract Wedderburn type alone does not retain.

2.2 Optional Lie/Hall Enrichment

An operator SOF does not automatically contain a Lie carrier. A **Lie/Hall enrichment** consists of an independently registered labelled family

$$X = \{X_g\}_{g\in G_0}, \qquad X_g^* = -X_g,$$

together with a declared filtration

$$\mathcal{H}_0 \subseteq \mathcal{H}_1 \subseteq \cdots.$$

Each $\mathcal{H}_d$ is a declared linear space of formal Lie expressions in the registered labels, and $H(X)$ denotes evaluation of $H \in \mathcal{H}_d$ on the family X. The filtration record must state whether d denotes Hall length, bracket depth, or a closure-generation round.

This registration must specify how X is obtained, including any logarithm branch, skew-adjoint projection, normalization, label map, and depth convention. An explicit induction rule from Y may provide such an enrichment, but $X = \mathrm{skew}(E_Y)$ is not sufficient unless the labelled selection rule is also declared. In particular, iQ_i and iI are not automatic Lie generators.

The layered SOF record is therefore one of:

$$\begin{aligned}
\text{operator core:}\quad & (V, Q, Y),\\
\text{operator-system layer:}\quad & (V, Q, Y, E_Y, S_{Q,Y}),\\
\text{positive-word closure:}\quad & (V, Q, Y, A_Y^+),\\
\text{observable star layer:}\quad & (V, Q, Y, A_Y^*),\\
\text{sector-star layer:}\quad & (V, Q, Y, A_{Q,Y}^*),\\
\text{Lie/Hall enrichment:}\quad & (V, Q, Y; X, \{\mathcal{H}_d\}_{d\geq 0}).
\end{aligned}$$

The layers are optional enrichments, not an assertion that all SOFs contain all of them.

2.3 Sectorization Origin

The source of the projectors is not part of the abstract SOF core. A sectorization may be:

- representation-derived, such as reducing or joint-spectral projectors;
- geometry-derived, such as block-diagonal Dirac or mesh/interface sectors;
- filtration-derived, such as reachable or communicating-state flags;
- graph-derived, such as vertex or color-class sectors;
- activation-derived, such as activation or rank regions;
- externally chosen, when a declared coarse coordinate system is supplied.

The origin must be recorded in a realization, because different origins can produce different admissibility domains and different observable families. Once the projectors and labelled alphabet are fixed, the static constructions below depend on the realized SOF data.

3 Declared Sectorized Realizations

3.1 No-Sector No-Shadow Principle

Let V carry a finite operator family Y. If no sectorization $\{Q_i\}$ is specified, then sector-indexed blocks and their typed shadows are not defined. If the only sectorization is the trivial one $\{I\}$, there are no distinct sector pairs and hence no nontrivial cross-sector support, routed bridge, or sector-indexed depth.

This is a definitional necessity statement, not a classification theorem. A global observable or an unsectorized diagnostic may still exist; it simply is not a sector-indexed SOF shadow.

3.2 Figure and Interface

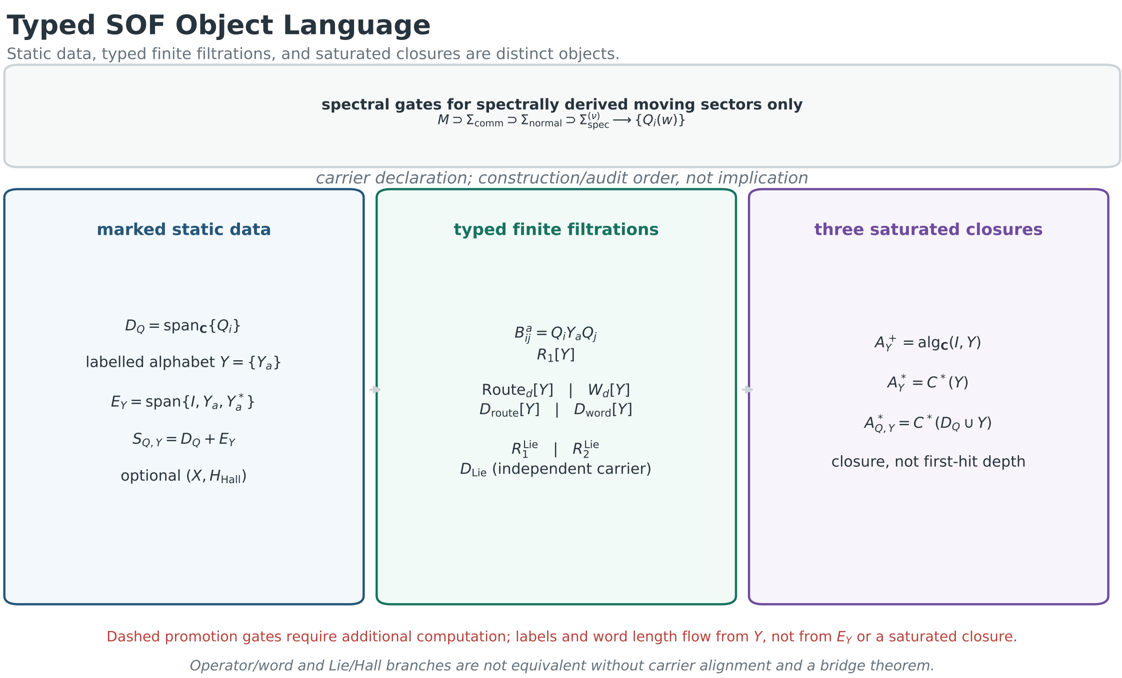

Figure 1. Typed SOF object language. The marked sector algebra, labelled operator alphabet, typed finite filtrations, and saturated closures are distinct layers. The operator/word and Lie/Hall branches are not identified without a bridge theorem.

3.3 Construction (Declared SOF Realization)

Let a finite source system provide:

1. a finite-dimensional complex Hilbert space V;
2. a compatible complete sectorization $\{Q_i\}_{i \in I}$;
3. a finite label set A and an observable extraction rule producing a possibly noninjective map $Y : A \to \mathcal{B}(V)$.

Then the data define an operator SOF core

$$\mathcal{F}_{\mathrm{op}} = (V, \{Q_i\}, Y : A \to \mathcal{B}(V)).$$

The realization is uniquely determined relative to the declared space, sectorization, extraction rule, label set, word convention, and normalization. If a Lie/Hall enrichment or a finite filtration is also supplied, the corresponding typed constructions are defined relative to those additional choices.

3.4 Well-Definedness

The extraction rule supplies the labelled map Y, and the compatible sectorization supplies the mutually orthogonal projectors. These are exactly the data of the operator SOF core. The definitions of D_Q, E_Y, $S_{Q,Y}$, and the three closure layers then use only the realized operators and the declared operative alphabet. A filtration or Lie/Hall carrier is available only when its additional registration data have been supplied. Thus, the construction is source-independent once these choices are fixed; however, it does not constitute a classification of source systems.

3.5 Realization Boundary

Sectorization may come from representation theory, geometry, a filtration, a graph, an activation, control/PDE data, or an external choice. All may enter the SOF language. The realization record must state whether the choice is canonical, constructed, truncated, or non-unique. Different realizations of the same source system need not be strictly equivalent.

The construction does not turn a source model into a unique SOF, and it does not promote a numerical sectorization to an exact one. Report-level or black-box uses belong to later diagnostic protocols and are not strict SOF objects by this construction alone.

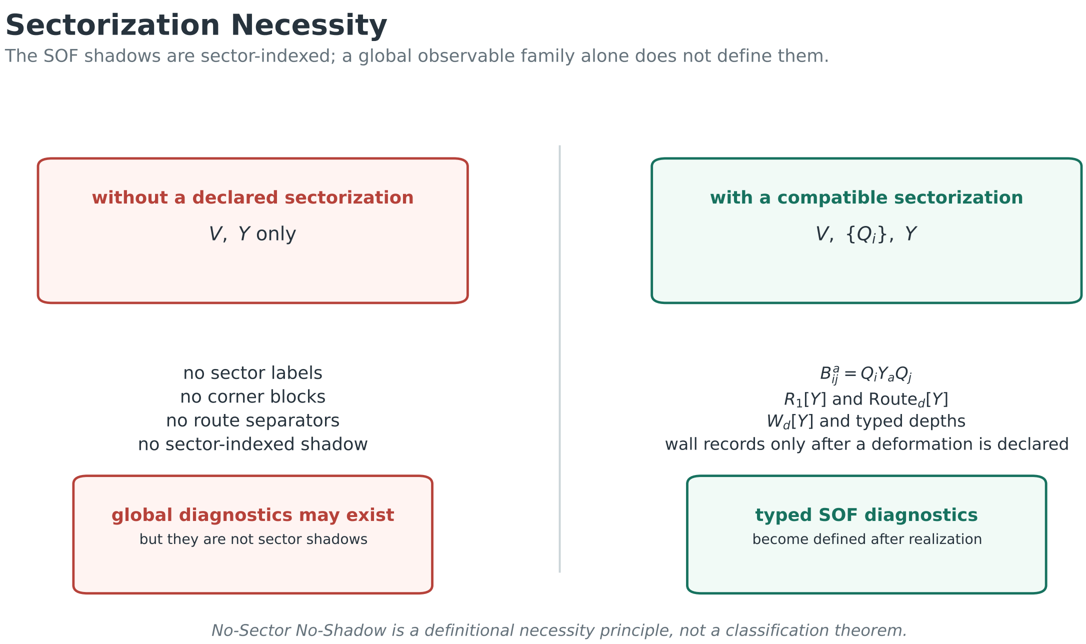

Figure 2. Sectorization necessity. Global observables exist without sectors, but sector-indexed blocks, routes, and typed shadows require the compatible sectorization.

4 Typed Static Constructions

4.1 Labelled Blocks and Direct Support

For $i, j \in I$ and $a \in A$, define the labelled block

$$B^a_{ij} = Q_i Y_a Q_j.$$

The labelled and aggregate direct supports are

$$R_{1,a}[Y](i,j) = \mathbf{1}[B^a_{ij} \neq 0], \qquad R_1[Y](i,j) = \max_{a \in A} R_{1,a}[Y](i,j).$$

The aggregate support forgets the witnessing label; the tensor $R_{1,a}[Y]$ retains it. The direction convention is

$$B^a_{ij} \neq 0 \quad \Longleftrightarrow \quad j \longrightarrow i.$$

For an off-diagonal pair $i \neq j$, the first operator corner

$$C^{(1)}_{ij} = Q_i E_Y Q_j$$

is an operator-system-derived corner space. For $i \neq j$, it need not itself be an operator system, nor is it a replacement for the generator-labelled block tensor. It agrees with aggregate direct support only when the selected alphabet is adjoint-closed with the declared direction convention. On the diagonal, $Q_i E_Y Q_i$ always contains $\mathbb{C} Q_i$ because $I \in E_Y$, so it cannot be identified with observable direct support.

4.2 Aggregate Support Paths

For $d \geq 1$, define the exact-length aggregate path shadow by

$$\mathrm{Path}_d(R_1[Y])(i,j) := \mathbf{1}\left[\begin{array}{c} \text{there exist } k_0 = j, k_1, \dots, k_d = i \\ \text{such that } R_1[Y](k_\ell, k_{\ell-1}) = 1 \\ \text{for every } 1 \leq \ell \leq d \end{array}\right].$$

Also set

$$\mathrm{Path}_0(R_1[Y])(i,j) = \delta_{ij}.$$

This is a path in the aggregate directed support graph, with direction $j \to i$. It has exactly d steps, forgets all generator labels, and imposes no compatibility condition between witnessing labels on adjacent edges. Repeated vertices are allowed, and a self-loop step is permitted exactly when the corresponding aggregate diagonal support $R_1[Y](k,k)$ is nonzero. These conventions are required for comparison with routed products that may contain nonzero diagonal sector blocks. Off-diagonal accessibility statements still restrict the endpoint pair to $i \neq j$.

4.3 Routed Products

For labels $a_1, \dots, a_d$ and intermediate sectors $\mathbf{k} = (k_1, \dots, k_{d-1})$, define

$$P^Y_{i,\mathbf{k},j}(a_d, \dots, a_1) = Q_i Y_{a_d} Q_{k_{d-1}} Y_{a_{d-1}} \cdots Q_{k_1} Y_{a_1} Q_j.$$

For $d = 1$, the intermediate tuple is empty and the routed product is $P^Y_{i,j}(a_1) = Q_i Y_{a_1} Q_j$. The corresponding routed-product space is

$$\mathcal{R}_{d,ij}[Y] := \operatorname{span}_{\mathbb{C}} \left\{ P^{Y}_{i,\mathbf{k},j}(a_d, \dots, a_1) : \mathbf{k} \in I^{d-1},\ (a_1, \dots, a_d) \in A^d \right\}.$$

The Boolean routed shadow is

$$\mathrm{Route}_d[Y](i,j) := \mathbf{1}[\mathcal{R}_{d,ij}[Y] \neq \{0\}].$$

Thus the shadow is nonzero exactly when at least one declared label tuple and intermediate-sector tuple produces a nonzero routed product. The space $\mathcal{R}_{d,ij}[Y]$ retains linear routed-product information; its Boolean shadow is not graph path closure.

4.4 Full Words

Use the same operative alphabet Y that defines $R_1[Y]$ and $\mathrm{Route}_d[Y]$. Define the exact-length word space

$$\mathcal{W}_d(Y) := \operatorname{span}_{\mathbb{C}}\{Y_{a_d} \cdots Y_{a_1} : (a_1, \dots, a_d) \in A^d\},$$

with $\mathcal{W}_0(Y) = \mathbb{C}I$, and its sector corner

$$\mathcal{W}_{d,ij}[Y] := Q_i \mathcal{W}_d(Y) Q_j.$$

The full-word shadow is

$$W_d[Y](i,j) := \mathbf{1}[\mathcal{W}_{d,ij}[Y] \neq \{0\}].$$

Completeness of the sectorization gives

$$Q_i Y_{a_d} \cdots Y_{a_1} Q_j = \sum_{\mathbf{k}} P^{Y}_{i,\mathbf{k},j}(a_d, \dots, a_1).$$

Consequently,

$$\mathcal{W}_{d,ij}[Y] \subseteq \mathcal{R}_{d,ij}[Y], \qquad W_d[Y] \subseteq \mathrm{Route}_d[Y] \subseteq \mathrm{Path}_d(R_1[Y]).$$

The reverse inclusions are not available in general. A full word may vanish by cancellation among routed terms, and a routed product may vanish by image–kernel alignment even when its Boolean path is present.

The dimensions of $\mathcal{W}_{d,ij}[Y]$ and $\mathcal{R}_{d,ij}[Y]$ are typed static data. Although later deformation theories may study their rank or dimension walls, this paper makes no moving-wall claim.

4.5 Typed Depths and Saturation

For exact extended-valued objects define

$$D_{\mathrm{route}}[Y](i,j) = \inf\{d \geq 1 : \mathrm{Route}_d[Y](i,j) = 1\},$$

$$D_{\mathrm{word}}[Y](i,j) = \inf\{d \geq 1 : W_d[Y](i,j) = 1\}.$$

Here $\inf \emptyset = \infty$. A finite computation reports $D^{(\leq d_{\max})} = d$ when it witnesses a first hit at $1 \leq d \leq d_{\max}$, and reports $D^{(\leq d_{\max})} =$ unreached otherwise. Promoting the latter value to exact mathematical infinity requires the relevant closure or saturation certificate. The sentinel 999 is never mathematical infinity.

The three saturated corner spaces are

$$A^{+}_{ij} := Q_i A^{+}_{Y} Q_j, \qquad A^{*}_{ij} := Q_i A^{*}_{Y} Q_j,$$

and

$$A_{ij}^{Q,*} := Q_i A_{Q,Y}^* Q_j.$$

Since

$$A_Y^+ = \operatorname{span}_{\mathbb{C}} \bigcup_{d\geq 0} \mathcal{W}_d(Y),$$

The condition $A_{ij}^+ \neq 0$ records positive-word saturation for the operative alphabet. In finite dimensions, the cumulative spaces $\sum_{\ell=0}^{d} \mathcal{W}_\ell(Y)$ stabilize at A_Y^+; however, the stabilized algebra alone does not retain the first stabilizing or first-hit length. The corner A_{ij}^* records star-word saturation and may use adjoints that are absent from a positive operative alphabet. The corner $A_{ij}^{Q,*}$ records sector-enriched star-saturation and may use projectors as internal route separators. In particular, a nonzero element such as

$$Q_i Y_a Q_k Y_b Q_j$$

may survive in $A_{ij}^{Q,*}$ even when the corresponding full-word corner $Q_i Y_a Y_b Q_j$, which sums over all intermediate routes, vanishes by cancellation. Thus positive-word, star-word, and routed star-saturation are not identified.

None of these corner spaces records the first-hit word length, a witnessing labelled word, or a witnessing route. For $i = j$, all three saturated corners are automatically nonzero because their defining unital algebras contain I, and hence

$$Q_i I Q_i = Q_i \in A_{ii}^+ \cap A_{ii}^* \cap A_{ii}^{Q,*}.$$

Saturated Boolean accessibility is therefore informative without qualification primarily on off-diagonal sector pairs. A nontrivial diagonal audit must remove the scalar sector identity. For any $A_{ii}^\bullet \in \{A_{ii}^+, A_{ii}^*, A_{ii}^{Q,*}\}$, one may use the Hilbert–Schmidt reduced space

$$\widetilde{A}_{ii}^\bullet := A_{ii}^\bullet \cap (\mathbb{C}Q_i)^{\perp_{\mathrm{HS}}},$$

or equivalently the vector-space quotient $A_{ii}^\bullet/\mathbb{C}Q_i$. The quotient is not asserted to be an algebra quotient. Registry or wall data must state whether a saturated-corner quantity is off-diagonal or uses this scalar-reduced diagonal convention. The condition $D_Q \subseteq A_Y^*$ is required before the observable star-closure and the sector-enriched star-closure coincide.

4.6 Lie/Hall Branch

Fix a total order on the Lie-label set G_0. For a registered Lie family $X = \{X_g\}_{g\in G_0}$, define

$$R_{1,g}^{\mathrm{Lie}}(i,j) = \mathbf{1}[Q_i X_g Q_j \neq 0], \qquad R_1^{\mathrm{Lie}}(i,j) = \max_g R_{1,g}^{\mathrm{Lie}}(i,j),$$

and

$$R_{2,g,h}^{\mathrm{Lie}}(i,j) = \mathbf{1}[Q_i [X_g, X_h] Q_j \neq 0], \qquad R_2^{\mathrm{Lie}}(i,j) = \max_{g<h} R_{2,g,h}^{\mathrm{Lie}}(i,j).$$

Here $\max \emptyset := 0$; equivalently, $R_2^{\mathrm{Lie}}(i,j) = 0$ when $|G_0| < 2$.

Given a declared Hall or Lie filtration $\{\mathcal{H}_d\}$,

$$D_{\mathrm{Lie}}(i,j) = \inf\{d : \exists H \in \mathcal{H}_d,\ Q_i H(X) Q_j \neq 0\}.$$

The depth index must state whether it means Hall length, bracket depth, or closure-generation round. Associative products $\mathrm{Route}_d[X]$ and $W_d[X]$ are allowed as diagnostics, but they are not Hall-filtered support and do not define D_{Lie}.

There is no canonical identification

$$R_1[Y] \equiv R_1^{\mathrm{Lie}}, \qquad D_{\mathrm{word}} \equiv D_{\mathrm{Lie}}.$$

Nor is $W_d[X]$ canonically identified with support at the corresponding Hall level. Each identification requires a separately declared bridge theorem and aligned registrations.

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5 Strict Morphisms and Functoriality

5.1 Operator Strict Morphisms

Let

$$\mathcal{F} = (V, Q, Y), \qquad \mathcal{F}' = (V', Q', Y')$$

be operator SOF cores. A **strict operator SOF morphism** $\Phi = (U, f, \phi) : \mathcal{F} \to \mathcal{F}'$ consists of:

1. an isometric embedding $U : V \to V'$;
2. an injective sector map $f : I \to I'$;
3. an injective operative-alphabet map $\phi : A \to A'$ that respects any explicitly registered adjoint labels;
4. a reducing-image condition for $P_f = \sum_{i \in I} Q'_{f(i)}$;
5. the intertwining identities

$$UQ_iU^* = Q'_{f(i)},$$

$$UY_aU^* = P_f Y'_{\phi(a)} P_f \qquad (a \in A).$$

The reducing-image condition is

$$[P_f, Y'_{\phi(a)}] = 0 \qquad (a \in A).$$

Completeness and the sector intertwining identities give

$$P_f = \sum_{i \in I} UQ_iU^* = UU^*.$$

The reducing-image condition prevents the matched observable from leaving the embedded selected sectors and returning through an untracked target sector. The matched-observable conjugation identity also intertwines adjoints.

The induced marked-sector map sends Q_i to $Q'_{f(i)}$, and the induced operator-system map sends E_Y into $P_f E_{Y'} P_f$. For the matched target family $\phi(Y) = \{Y'_{\phi(a)} : a \in A\}$, conjugation by U yields an algebra isomorphism and a unital $*$-isomorphism, respectively:

$$\mathrm{Ad}_U : A_Y^+ \xrightarrow{\cong} P_f A^{+\prime}_{\phi(Y)} P_f, \qquad \mathrm{Ad}_U : A_Y^* \xrightarrow{\cong} P_f A^{*\prime}_{\phi(Y)} P_f.$$

Here the primed closures are formed in $\mathcal{B}(V')$ from the matched target family only. Because P_f reduces every matched observable, both compressed closures are algebras on $P_f V'$ with unit P_f.

For the sector-enriched layer, define the **matched sector-star closure**

$$A^{*\prime}_{f(Q),\phi(Y)} := C^*_{P_f\mathcal{B}(V')P_f}\left(\{Q'_{f(i)} : i \in I\} \cup \{P_f Y'_{\phi(a)} P_f : a \in A\}\right).$$

Conjugation by U then gives the unital $*$-isomorphism

$$\mathrm{Ad}_U : A^*_{Q,Y} \xrightarrow{\cong} A^{*\prime}_{f(Q),\phi(Y)}.$$

This matched closure satisfies

$$A^{*\prime}_{f(Q),\phi(Y)} \subseteq P_f A^*_{Q',Y'} P_f.$$

Because $P_f \in D_{Q'} \subseteq A^*_{Q',Y'}$, the right-hand side is a C^*-corner rather than merely a linear corner. Equality with the full target corner is not asserted, because unmatched target observables may contribute additional corner elements.

A strict equivalence is a strict morphism for which U is unitary and f and ϕ are bijections. The optional Lie/Hall enrichment adds an injective label map $\psi : G_0 \to G'_0$. It must satisfy

$$[P_f, X'_{\psi(g)}] = 0, \qquad U X_g U^* = P_f X'_{\psi(g)} P_f.$$

Let ψ_* denote relabelling of formal Lie expressions. Filtration preservation means

$$\psi_*(\mathcal{H}_d) \subseteq \mathcal{H}'_d \qquad \text{for every } d,$$

and, for every $H \in \mathcal{H}_d$,

$$U H(X) U^* = P_f (\psi_* H)(X') P_f.$$

A strict Lie/Hall equivalence additionally requires ψ to be bijective and the filtration-preservation condition to hold in both directions.

5.2 Strict-Category Proposition

Proposition 5.1 (Closure of Strict Morphisms)**.**
Finite operator SOF cores and strict operator morphisms form a category, denoted

$$\mathsf{SOF}_{\mathrm{op,str}}.$$

Lie/Hall-enriched SOFs and filtration-preserving strict morphisms form a separate category over the operator category, denoted

$$\mathsf{SOF}_{\mathrm{Lie,str}}.$$

Proof. *Identity isometries and identity label maps satisfy all defining conditions. For composition, let*

$$\Phi = (U, f, \phi) : \mathcal{F} \longrightarrow \mathcal{F}', \qquad \Phi' = (U', f', \phi') : \mathcal{F}' \longrightarrow \mathcal{F}''.$$

The composite isometry and label maps are $U'U$, $f' \circ f$, and $\phi' \circ \phi$. The corresponding selected-sector projection is

$$P_{f' \circ f} = \sum_{i \in I} Q''_{f'(f(i))} = U' P_f U'^*.$$

Write $P_{f'} = U'U'^*$ and $Z = Y''_{\phi'(\phi(a))}$. Since $P_{f'}$ reduces Z and $U'^* Z U' = Y'_{\phi(a)}$, the first reducing condition gives

$$
\begin{aligned}
P_{f'\circ f} Z &= U' P_f Y'_{\phi(a)} U'^* \\
&= U' Y'_{\phi(a)} P_f U'^* = Z P_{f'\circ f}.
\end{aligned}
$$

Moreover,

$$
\begin{aligned}
(U'U) Y_a (U'U)^* &= U' P_f Y'_{\phi(a)} P_f U'^* \\
&= P_{f'\circ f} Y''_{\phi'(\phi(a))} P_{f'\circ f}.
\end{aligned}
$$

The sector identities compose in the same way, so the operator composite is strict. For Lie/Hall enrichments, the composite relabelling is $(\psi' \circ \psi)_* = \psi'_* \circ \psi_*$. It preserves every filtration level, and the same calculation proves the reducing and intertwining identities for each evaluated formal Lie expression. Thus identities and composition are closed in both categories. Associativity is inherited from composition of isometries and label maps.

The existence of a Lie/Hall carrier is not inferred from an operator morphism.

5.3 Operator/Word Functoriality

Theorem 5.2 (Operator/Word Support Preservation and Depth Monotonicity)**.**

Let $\Phi = (U, f, \phi) : \mathcal{F} \to \mathcal{F}'$ be a strict operator SOF morphism. Then for all matched labels and sectors:

$$
B^a_{ij} \neq 0 \iff Q'_{f(i)} Y'_{\phi(a)} Q'_{f(j)} \neq 0.
$$

The same equivalence holds for every matched routed product and every matched full word. At the linear-space level,

$$
\mathrm{Ad}_U(\mathcal{R}_{d,ij}[Y]) \subseteq \mathcal{R}'_{d,f(i)f(j)}[Y'],
$$

and

$$
\mathrm{Ad}_U(\mathcal{W}_{d,ij}[Y]) \subseteq \mathcal{W}'_{d,f(i)f(j)}[Y'].
$$

Therefore

$$
R_1[Y](i,j) \Longrightarrow R_1[Y'](f(i), f(j)),
$$

$$
\mathrm{Route}_d[Y](i,j) \Longrightarrow \mathrm{Route}_d[Y'](f(i), f(j)),
$$

and

$$
W_d[Y](i,j) \Longrightarrow W_d[Y'](f(i), f(j)).
$$

Taking the first-hit infima gives

$$
D'_{\mathrm{route}}(f(i), f(j)) \leq D_{\mathrm{route}}(i,j), \qquad D'_{\mathrm{word}}(f(i), f(j)) \leq D_{\mathrm{word}}(i,j)
$$

whenever the source depths are finite. Under strict equivalence, the corresponding supports and depths are equal.

Proof. *The sector and observable intertwining identities give*

$$
U Q_i Y_a Q_j U^* = Q'_{f(i)} Y'_{\phi(a)} Q'_{f(j)}.
$$

Because P_f reduces every matched observable, routed products satisfy

$$\begin{aligned}&UQ_iY_{a_d}Q_{k_{d-1}}\cdots Q_{k_1}Y_{a_1}Q_jU^*\\&\qquad=Q'_{f(i)}Y'_{\phi(a_d)}Q'_{f(k_{d-1})}\cdots Q'_{f(k_1)}Y'_{\phi(a_1)}Q'_{f(j)},\end{aligned}$$

and full words satisfy

$$UQ_iY_{a_d}\cdots Y_{a_1}Q_jU^*=Q'_{f(i)}Y'_{\phi(a_d)}\cdots Y'_{\phi(a_1)}Q'_{f(j)}.$$

Because U is an isometry, nonzero source matrices remain nonzero. Consequently, a source filtration witness supplies a target witness of no greater depth. A strict equivalence provides the reverse identities.

5.4 Lie/Hall Functoriality

Theorem 5.3 (Lie/Hall Support Preservation and Depth Monotonicity)**.**
Let Φ be a strict morphism between Lie/Hall-enriched SOFs with the filtration-preserving Lie-label map declared above. Then

$$R_1^{\mathrm{Lie}}(i,j)\Longrightarrow R_1^{\mathrm{Lie}\prime}(f(i),f(j)),$$

$$R_2^{\mathrm{Lie}}(i,j)\Longrightarrow R_2^{\mathrm{Lie}\prime}(f(i),f(j)),$$

and

$$D'_{\mathrm{Lie}}(f(i),f(j))\le D_{\mathrm{Lie}}(i,j)$$

whenever the source depth is finite. Under strict Lie/Hall equivalence, the matched supports and depths are equal.

Proof. *The carrier intertwining gives*

$$UX_gU^*=P_fX'_{\psi(g)}P_f.$$

Consequently, conjugation by U intertwines every matched commutator. The formal relabelling ψ_* likewise matches every filtered Lie expression without increasing its filtration index. Therefore, the same nonzero-block and filtration-witness argument used in the operator/word functoriality theorem proves the claims.

5.5 Functoriality Boundary

The theorems are functorial support and depth statements, not completion results. They do not imply that direct support determines routed composition, that routed products determine full words, or that low-order Lie support determines Lie depth. Matched multiplication closures are carried by exact isomorphisms, whereas support in the full target alphabet is preserved by inclusion and first-hit depth is only non-increasing. Strict morphisms also do not create a deformation morphism. A generator-weight path, a state-mixing path, or a training trajectory may fail to be isometric, label-preserving, or reducing.

A deformation analysis may use a weaker category, provisionally denoted $\mathsf{SOF}_{\mathrm{def}}$, but this paper does not define its arrows.

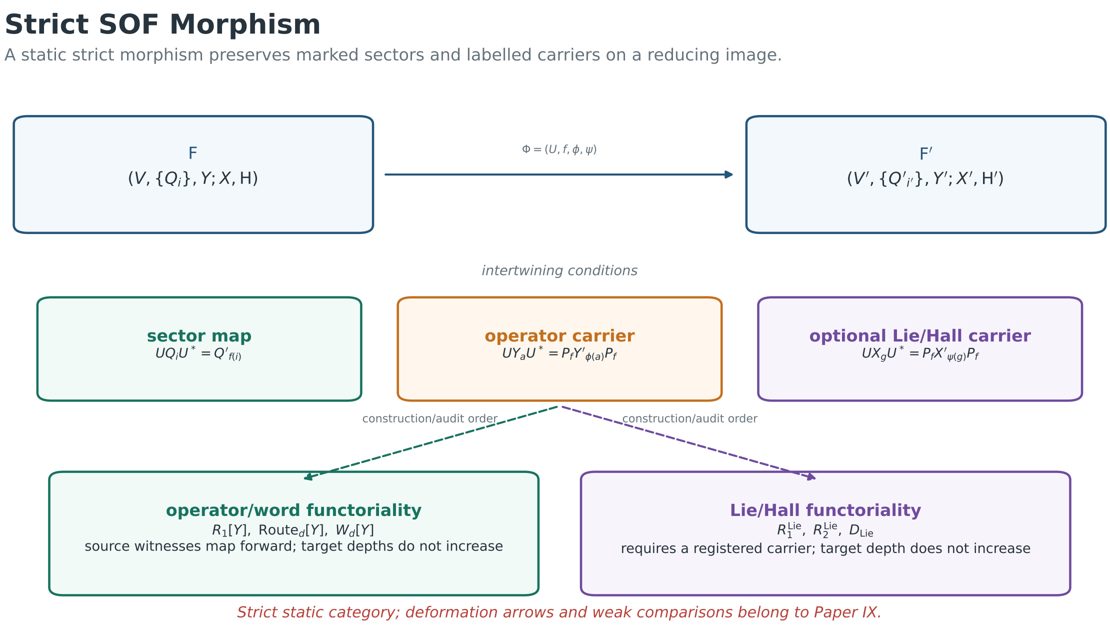

Figure 3. Strict SOF morphism. The operator/word and optional Lie/Hall carriers have separate label maps and separate functorial outputs. The static morphism is not a deformation arrow.

6 Exact Marked Finite Permutation Realizations

The abstract static interface does not require a group action. Finite permutation actions nevertheless provide a useful exact conformance family: the Hilbert space, projectors, represented letters, routed products, and words all admit finite combinatorial descriptions, while the marked partition and source labels remain visible.

6.1 Labelled Action Data

Let a finite group G act on a finite set Ω, let A be a finite nonempty label set, and let

$$y : A \longrightarrow G$$

be a declared source-letter map. Neither y nor the represented map below is assumed injective. The permutation representation on $V = \mathbb{C}^{\Omega}$ is

$$\rho(g)e_\omega = e_{g\omega}, \qquad Y_a = \rho(y(a)).$$

Let

$$\Pi = \{\Omega_i\}_{i\in I}, \qquad \Omega = \bigsqcup_{i\in I} \Omega_i,$$

be a declared marked partition. Define the coordinate projector

$$Q_i^{\Pi} e_\omega = \begin{cases} e_\omega, & \omega \in \Omega_i, \\ 0, & \omega \notin \Omega_i. \end{cases}$$

Here G and y retain source provenance, while the static SOF core consumes the represented labelled map and marked partition. Neither the action nor the label map is required to be faithful.

Proposition 6.1 (Marked Finite Permutation Realization)**.**
The data $(\Omega, G, \rho, A, y, \Pi)$ *define an exact operator SOF core*

$$\mathcal{F}_{\rho,y,\Pi} = (\mathbb{C}^{\Omega}, \{Q_i^{\Pi}\}_{i\in I}, (Y_a)_{a\in A}).$$

The operative alphabet is the labelled map $Y : A \to U(\mathbb{C}^{\Omega})$*, not its image set. Thus* $a \neq b$ *and* $Y_a = Y_b$ *are compatible with the realization. The action and labelled alphabet do not by themselves select* Π*; different marked partitions of the same represented action define different marked SOF data unless an additional equivalence is supplied.*

Proof. *Every* Y_a *is a permutation unitary. The coordinate projectors satisfy*

$$(Q_i^{\Pi})^* = Q_i^{\Pi}, \qquad Q_i^{\Pi} Q_j^{\Pi} = \delta_{ij} Q_i^{\Pi}, \qquad \sum_i Q_i^{\Pi} = I.$$

They act on the same space as the labelled represented family, so the defining conditions of an operator SOF core hold exactly. Nothing in ρ *selects a partition as part of the SOF data, and replacing* $Y : A \to U(V)$ *by its image would identify distinct labels whenever the represented map is noninjective.*

This gives the ***Marked-Realization Principle****:*

$$\textit{finite action} + \textit{ labelled operative map} + \textit{ marked partition} \longrightarrow \textit{exact finite SOF realization.}$$

The arrow is a construction. It does not assert a canonical partition or a classification of finite representations.

6.2 Deterministic Route/Word Coincidence

Coordinate partitions and permutation letters have a special property that does not hold for general operator SOFs.

Proposition 6.2 (Permutation Route/Word Coincidence)**.**
For a marked finite permutation realization and every $d \geq 1$*,*

$$\mathrm{Route}_d[Y] = W_d[Y].$$

This is equality of Boolean support relations. It does not identify the linear spaces $\mathcal{R}_{d,ij}[Y]$ *and* $\mathcal{W}_{d,ij}[Y]$*.*

Proof. *Fix a labelled word and a source basis state. Because each letter is a permutation, the state follows one unique sequence of intermediate states and therefore one unique sequence of marked sectors. A nonzero full word corner supplies that routed witness. Conversely, a nonzero routed product contains a source basis state whose unique permutation trajectory follows the declared intermediate sectors and ends in the target sector; the corresponding full-word corner is therefore nonzero. Taking the union over labelled words and intermediate-sector tuples proves the equality.*

The proposition is carrier-specific. It removes route cancellation inside this exact realization family, but it does not identify either relation with aggregate Boolean graph paths.

6.3 Exact Finite Word Saturation

For $d \geq 1$, let

$$C_{\leq d}[Y] = \left\{ Y_{a_k} \cdots Y_{a_1} : 1 \leq k \leq d, (a_1, \dots, a_k) \in A^k \right\}.$$

This is a cumulative represented-operator set. It is distinct from the exact-length word space $\mathcal{W}_d(Y)$ and from the set of labelled words: different labelled words may evaluate to the same represented permutation.

Proposition 6.3 (Exact Finite Positive-Word Saturation)**.**
For every marked finite permutation realization, there is a finite $d_{\text{sat}} \geq 1$ *such that*

$$C_{\leq d_{\text{sat}}}[Y] = C_{\leq d_{\text{sat}}+1}[Y] = \langle Y(A) \rangle,$$

where $\langle Y(A) \rangle$ *is the finite represented subgroup generated by the operative letters. Hence, for every sector pair* (i, j)*, exhaustive labelled breadth-first search gives either an exact shortest nonempty first-hit word or an exact certificate that* $D_{\text{word}}[Y](i, j) = \infty$*.*

Proof. *All represented words lie in the finite permutation group on* Ω*, so the increasing sequence* $C_{\leq d}[Y]$ *stabilizes. Because* A *is nonempty and each represented letter has finite order, the identity occurs at a positive length. At stabilization the cumulative set is closed under right multiplication by every declared letter, so no later positive word can add a represented operator. Conversely, every generated represented operator has a finite positive-word witness. Exhaustive labelled breadth-first search therefore records the minimum positive length of every represented operator. Testing all saturated operators against* $Q_i(\cdot)Q_j$ *decides each sector pair exactly.*

A saturation receipt must bind the label alphabet, the label-to-operator map, the marked partition, cumulative closure sizes, stabilization depth, right-multiplication closure, the shortest nonempty identity word, and the first-hit witnesses. A bounded search without this closure evidence still reports `unreached`*, not infinity.*

6.4 Four-State Exact Separation Witness

Let

$$\Omega = \{0, 1, 2, 3\}, \qquad \Pi = \{\{0, 1, 2\}, \{3\}\},$$

and declare two labels represented by

$$Y_a = (2\ 3), \qquad Y_b = (0\ 1)(2\ 3).$$

Proposition 6.4 (Boolean Path Overestimate)**.**
With rows indexed by target sector and columns by source sector,

$$R_1[Y] = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{Path}_2(R_1[Y]) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

whereas

$$\mathrm{Route}_2[Y] = W_2[Y] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus

$$W_2[Y] = \mathrm{Route}_2[Y] \subsetneq \mathrm{Path}_2(R_1[Y]).$$

Proof. *The direct-support matrix follows by applying the two permutations to the two marked sectors. Boolean squaring gives the displayed full* Path_2 *relation. The four labelled words of length two evaluate to either* I *or* $(0\ 1)$ *because*

$$Y_a^2 = Y_b^2 = I, \qquad Y_a Y_b = Y_b Y_a = (0\ 1).$$

Both represented operators preserve the two marked sectors, so $W_2[Y]$ *is diagonal. Proposition 6.2 gives the same relation for* $\mathrm{Route}_2[Y]$*.*

All data in this witness are exact: no tolerance, approximate rank, or asymptotic enters. Thus aggregate support composition can overestimate actual words even for permutation operators on a complete finite coordinate partition.

6.5 Promoted Conformance Certificate

The paper-owned certificate promotes only the static realization and word-filtration claims needed here. Its accepted scope is:

Family or control	Declared scope	Marked sectorization	Promoted role
modular permutation actions	$\mathbb{P}^1(\mathbb{F}_p)$ for $p = 3, 5, 7, 11, 13, 17$	orbits of the declared T operator	six exact conformance records
triangle-group permutation actions	signatures $(2, 3, 7)$, $(2, 4, 5)$, $(3, 3, 4)$ through index 7	separate x-, y-, and z-cycle partitions	seventeen actions and fifty-one marked realizations
four-state control	two labelled permutations on four states	one declared two-sector partition	strict Boolean-path overestimate witness

Among the seventeen triangle records, twelve retain all three declared signature orders and five are explicitly marked as proper-order-divisor quotients. The bounded census contains three pairs of distinct labels with equal represented permutation operators. All fifty-seven modular/triangle marked realization–sectorization records carry complete finite positive-closure receipts; no sector pair in those records remains unreachable after saturation. The four-state hostile control carries its own exact first-hit and finite-closure replay record.

The paper-owned conformance artifact and its validation receipt are source-addressed in Appendix A. Their artifact IDs are `P8V2.1-CONFORMANCE` and `P8V2.1-REPLAY`. The two exploratory source bundles remain provenance inputs; they do not acquire Paper VIII evidence status independently of the paper-owned promotion artifact and receipt.

This evidence is a **Computational Certificate**. The propositions above are proved from the declared finite-action hypotheses and do not depend on the census counts. Graph-Laplacian, spanning-tree, surface, Hecke, moduli, Selberg, and automorphic fields are outside the promoted claim surface.

---

7 Other Realization Classes

The following examples illustrate the broader source boundary. Except for the promoted finite-permutation family above, they are not new computational evidence for the theorem layer developed here.

The static object can be interpreted as a marked geometry of coarse-grained information accessibility. The sectorization supplies source-dependent coarse coordinates; labelled operator blocks, routed products, words, and any independently registered Lie/Hall fields describe different forms of cross-sector propagation. This interpretation does not identify their carriers or make the sectorization canonical.

7.1 Representation-Derived Sectors

Finite-group representations may supply invariant blocks or joint-spectral projectors. The exact family above instead uses separately declared marked state partitions and therefore does not claim that a representation selects its own canonical SOF sectors. Rubik QT/HT sectors provide a motivating realization, while their exact numerical status remains owned by the relevant earlier papers.

7.2 Geometry-Derived Sectors

Finite spectral triples may use block-diagonal Dirac operators to define sectors. Mesh/interface partitions and other geometric decompositions provide additional examples. The sectorization need not originate in irreducible representation theory.

7.3 State, Graph, and Activation Sectors

Finite Markov systems may use communicating classes or state flags. Graph systems may use vertex, edge, color, or spectral sectors. Neural systems may use activation or rank regions. In each case the observable family and the sector provenance must be declared before a typed SOF audit is meaningful.

7.4 External Sectorizations

An externally chosen coarse coordinate system is admissible when the projectors, completeness, normalization, and observable family are explicit. Such a choice is not automatically canonical or source-invariant.

---

8 Claim Status and Boundary

The No-Sector No-Shadow Principle is a definitional structural principle. Definitions are not evidence-level claims, so it is not assigned one of the four statuses in the table.

Background facts used here include finite-dimensional C^*-algebra semisimplicity and Wedderburn decomposition. They are standard results, not reader-facing claims or contributions of this paper.

Claim	Status
strict operator category and carrier-qualified Lie/Hall category	Theorem
operator/word support preservation and depth monotonicity	Theorem
Lie/Hall support preservation and depth monotonicity	Theorem
marked finite permutation realization	Theorem
permutation route/word coincidence	Theorem
exact finite positive-word saturation	Theorem
four-state Boolean path overestimate	Theorem
modular and bounded triangle-group conformance census	Computational Certificate
source-specific realization and equivalence criteria	Research Program
typed bridge criteria between operator/word and Lie/Hall branches	Research Program
weak and deformation morphism structures	Research Program
conditional low-order promotion and completion criteria	Research Program

8.1 What This Paper Does Not Claim

This paper does not claim:

1. a classification of all source systems admitting a SOF realization;
2. uniqueness of compatible sectorization;
3. a universal wall or deformation theory;
4. an unconditional relation between word depth and Lie depth;

5. a complete weak morphism category;
6. new numerical evidence for Rubik, quantum, Markov, graph, neural, or other application species outside the promoted exact finite-permutation family;
7. a canonical sectorization determined by a finite representation;
8. a Lie/Hall carrier induced from the promoted permutation letters;
9. generic strict separation of A_Y^+ and A_Y^* from finite-group data;
10. a surface, Hecke, moduli, Selberg, or automorphic interpretation of the finite Schreier diagnostics.

The stable claim is narrower: once a compatible sectorization, a labelled observable map, and any required filtration or Lie/Hall enrichment have been declared, the resulting static typed constructions admit a strict object language and carrier-qualified functoriality statements. Finite permutation actions provide one exact marked conformance family, not a replacement for the abstract object language.

---

9 Conclusion

This paper fixes the static typed SOF object language. The marked sectorization, possibly noninjective labelled operative map, observable operator system, and three multiplication closures remain distinct data. Routed products and full words form one finite-filtration branch; an optional Lie/Hall carrier forms an independently registered branch. Strict morphisms carry matched multiplication closures by exact algebraic or $*$-isomorphisms, preserve matched witnesses, and make target first-hit depths non-increasing, with equality reserved for strict equivalence. For finite coordinate-sector permutation realizations, routed and full-word support coincide, finite represented-image saturation decides exact first-hit word depth, and aggregate Boolean paths may nevertheless strictly overestimate actual words.

These results concern static objects and carrier-qualified functoriality. The paper-owned modular, triangle-group, and four-state certificates are exact conformance witnesses; they do not serve as premises for the theorem layer. No deformation field, wall classification, compiler contract, or downstream application report is introduced.

---

10 Outlook

Paper IX takes a separately supplied object trajectory and studies its typed SOF observation/deformation record. The underlying dynamics, parameter update, or intervention supplies each object-state transition; observed projectors, observables, and derived fields may vary along that trajectory. Wall loci are typed discriminants of the observation/deformation record over a declared admissible domain. Paper VI supplies only a normality-gated spectral interface and pointwise registrations; it is not a positive moving SOF theorem.

Paper X studies capability-aware compilation and Registry evidence through the pipeline

$$\begin{aligned}\text{source and admission} &\longrightarrow \text{Capability Manifest} \longrightarrow \text{Typed SOF IR}\\ &\longrightarrow \text{Report Profile} \longrightarrow \text{supported report claims.}\end{aligned}$$

The strict-admission branch factors through the static realization construction of the present paper. The resulting pipeline is neither a universal accessibility ladder nor a universal dynamics theorem. The Capability Manifest, Typed SOF IR, and Report Profile belong to Paper X rather than to the static object defined here.

Open promotion problems include proxy/shadow bridges, route/word criteria beyond deterministic permutation carriers, and saturation certificates for nonfinite or numerically represented operator families. Word/Lie comparisons and weaker comparison morphisms are also open. These are future typed results, not implicit consequences of the static category.

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11 Related Work and Novelty Boundary

Finite-dimensional representation theory and Wedderburn–Artin decomposition provide the ambient structural background [5, 10, 6]. The finite-dimensional C^*-algebra and operator-system terminology is standard [8, 9]. The category language is standard mathematical bookkeeping [7].

Papers IV–VII provide independent compatible interfaces: fixed spectral arrangements, fixed typed accessibility objects, normality-gated linearized registrations, and projected-composition incidence [2, 1, 4, 3]. They are not premises that promote one typed carrier into another. In particular, this paper does not identify a routed product with a full word or a commutator with Lie depth.

The contribution here is the marked static SOF object language: sector projectors remain marked, observable labels remain visible even when the represented operator map is noninjective, finite filtrations remain separate from positive, star, and sector-enriched closures, and optional Lie/Hall data are independently registered. The exact finite-permutation family is a conformance witness for that interface and a strict control against replacing actual ordered words by aggregate Boolean graph powers.

Appendix A Computational Artifacts

The following repository artifacts support the finite-permutation conformance certificate. The default directory is `experiments/paper8/`; paths are relative to that directory.

Artifact	Role	Short path
A1	exact modular finite-carrier source bundle	`../exploratory/carrier_realizations/fuchsian_schreier/results/modular_p1_census_v2.json`
A2	exact bounded triangle-group source bundle	`../exploratory/carrier_realizations/fuchsian_schreier/results/triangle_low_index_census_v2.json`
A3	paper-owned promotion and saturation replay	`validation/promote_marked_finite_realizations_v2_1.py`
A4	conformance artifact, ID `P8V2.1-CONFORMANCE`	`results/v2.1/marked_finite_realization_conformance_v2_1.json`
A5	fail-closed paper-owned replay validator	`validation/validate_marked_finite_realizations_v2_1.py`
A6	validation receipt, ID `P8V2.1-REPLAY`	`results/v2.1/marked_finite_realization_conformance_v2_1.validation-receipt.json`
A7	human-readable projection of A4	`results/v2.1/marked_finite_realization_conformance_v2_1.md`

All listed artifacts are available in the RIME repository.

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